

Virtual LAN (VLAN)

A Virtual local area network is a logical way to split a single physical network into multiple separate virtual networks.

- It reduces unnecessary broadcast traffic, enhances security, and improves network performance, flexibility, and manageability.
- Traditionally, broadcast domains were separated using Layer 3 devices (routers).
- **Same VLAN:** Devices communicate directly within the same broadcast domain.
- **Different VLANs:** Communication requires Inter-VLAN Routing via a router or a Layer 3 switch (SVI).

VLAN ID Ranges

- **VLAN 0 and VLAN 4095:** Reserved VLAN IDs are defined by IEEE 802.1Q and used internally for protocol operations and cannot be configured or assigned to ports.
- **VLAN 1 (Default VLAN):** The default VLAN on Cisco switches and all switch ports are assigned to VLAN 1 by default. It cannot be deleted and is commonly used for control and management protocols such as STP, CDP, and VTP.
- **VLAN 2 to VLAN 1001 (Normal VLAN Range):** This range is used for regular VLAN configuration. VLANs in this range are fully configurable, editable, and deletable, and are stored in the switch's VLAN database.
- **VLAN 1002 to VLAN 1005:** Reserved VLANs for legacy network technologies such as FDDI and Token Ring. These VLANs are predefined and cannot be removed, even if the technologies are no longer in use.
- **VLAN 1006 to VLAN 4094 (Extended VLAN Range):** Designed for large-scale networks requiring a high number of VLANs. VLANs in this range are stored in the **running configuration** and typically require the switch to operate in VTP transparent mode.

Types of VLAN Links

- **Access Link:** Connects a VLAN-unaware end device (such as a PC) to a VLAN-aware switch. Frames transmitted over an access link are untagged and belong to a single VLAN.

enable

configure terminal

vlan 10

name Sales

exit

- **Trunk Link:** Connects VLAN-aware devices such as switch-to-switch or switch-to-router links. It carries traffic from multiple VLANs using IEEE 802.1Q tagging.

configure terminal

interface GigabitEthernet 0/1

switchport mode trunk

#Specify Allowed VLANs

switchport trunk allowed vlan 10,20,30

end

- **Hybrid Link:** Supports both tagged and untagged frames on the same link. It is typically vendor-specific and not part of the IEEE 802.1Q standard.

Advantages

- **Improved Security:** Sensitive traffic is isolated within separate VLANs.
- **Enhanced Performance:** Limits broadcast and multicast traffic, reducing unnecessary network load.
- **Simplified Management:** Logical grouping of users or departments (e.g., HR, Finance) simplifies administration.
- **Flexibility:** Devices can be moved between VLANs without physical cable changes.
- **Cost Efficiency:** Reduces the need for multiple physical networks and excessive routing hardware.
- **Scalability:** Large networks can be divided into manageable segments.

Disadvantages

- **Increased Configuration Complexity:** Requires careful planning and configuration.
- **VLAN ID Limitations:** Standard VLAN IDs are limited, which can affect very large networks.
- **Security Risks:** Misconfigurations may allow VLAN hopping attacks.
- **Interoperability Issues:** Compatibility problems may arise with non-standard or legacy devices.
- **Troubleshooting Difficulty:** Isolated traffic flows can make fault diagnosis more complex.

VLAN Hopping

VLAN Hopping is an attack where the attacker is able to send traffic from one VLAN into another.

IEEE 802.1Q

IEEE 802.1Q is the networking standard that supports Virtual LANs (VLANs) on an Ethernet network. When you have a trunk link connecting two switches, or a switch and a router, that single physical connection needs a way to carry traffic for multiple VLANs while keeping them logically separated.

How 802.1Q Works: The 4-Byte Tag

802.1Q accomplishes VLAN separation by inserting a 4-byte tag into the original Ethernet frame header. This tag is inserted into the frame right after the Source MAC Address and before the original EtherType/Length field.

Because this tag is injected into the middle of an existing frame, the frame's overall size increases, and its contents change. Consequently, the switch must recalculate the Frame Check Sequence (FCS) trailer at the end of the frame so the receiving device doesn't drop it as a corrupted packet.

Native VLAN

The major exception to this tagging process is the **Native VLAN**.

- By default, the Native VLAN is VLAN 1.
- Traffic belonging to the Native VLAN is sent completely **untagged** across the trunk link. The switch does not insert the 4-byte 802.1Q header into these frames.
- When a receiving switch gets an untagged frame on an 802.1Q trunk port, it automatically forwards that frame into whatever VLAN is configured as the Native VLAN on that specific interface.

```
configure terminal
interface GigabitEthernet 0/1
switchport trunk native vlan 99
end
```

End Devices (Access Ports)

Standard network devices—like desktop computers, laptops, and typical printers—do not understand VLAN tags.

- When a PC sends data, it sends **untagged traffic**.
- The switch port connected to that PC is configured as an "access port" assigned to a specific VLAN (e.g., VLAN 10).
- When the switch receives the untagged frame from the PC, the switch internally tags it with VLAN 10 so it knows how to route it through the network.
- Before the switch sends a frame back out to the destination PC, it strips the tag off, delivering an **untagged frame** to the device.

Switch-to-Switch (Trunk Ports and the Native VLAN)

When traffic travels between two switches over a trunk link, it usually carries data for multiple VLANs. The switch must add a 4-byte 802.1Q tag to the frames so the receiving switch knows which VLAN the traffic belongs to.

However, there is one exception on a trunk link: the **Native VLAN**.

- You configure one VLAN on the trunk to be the Native VLAN (default is VLAN 1).
- When a switch needs to send traffic that belongs to the Native VLAN across the trunk, it deliberately **does not add a tag**. It sends it as **untagged traffic**.
- When the receiving switch gets an untagged frame on its trunk port, it assumes that frame belongs to its locally configured Native VLAN and processes it accordingly.

Note:

Untagged Traffic: A normal network frame without a VLAN ID.

Untagged VLAN (Native VLAN): The specific VLAN on a trunk link where the switches agree to send traffic without a tag.

Troubleshooting Command list

General VLAN Commands

- show vlan brief
- show vlan id <vlan-id>
- show mac address-table vlan <vlan-id>
- show spanning-tree vlan <vlan-id>
-

Access Port Commands

- show interfaces status
- show interfaces <interface-id> switchport
- show mac address-table interface <interface-id>
- show running-config interface <interface-id>

Trunk Port Commands

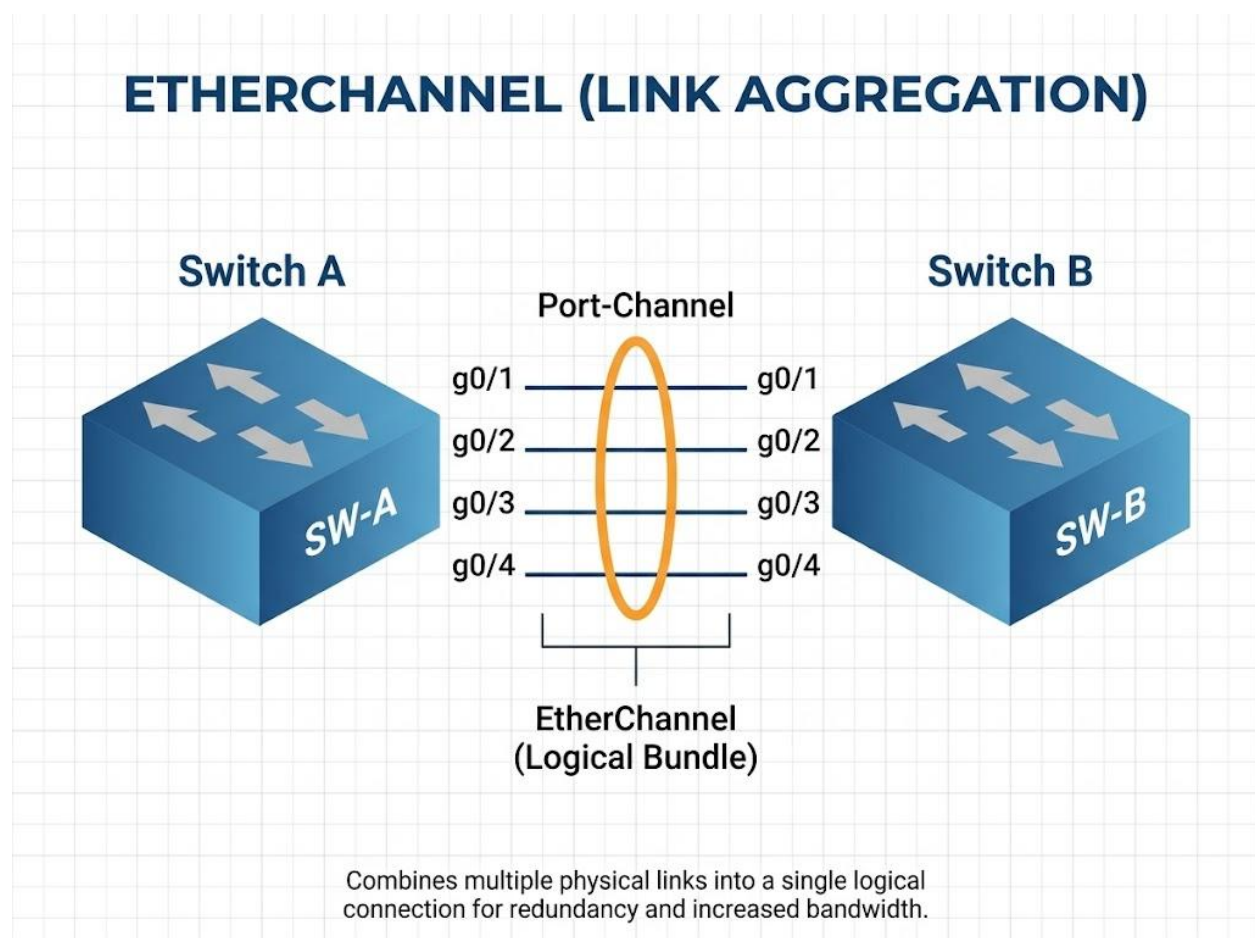
- show interfaces trunk
- show interfaces <interface-id> trunk
- show interfaces <interface-id> switchport
- show cdp neighbors
- show cdp neighbors detail

General Commands

- show logging
- show errdisable recovery
- show interfaces <interface-id> counters errors

EtherChannel

EtherChannel is a port aggregation technology that groups multiple physical Ethernet links into a single logical interface. This provides fault tolerance and scales bandwidth without causing Spanning Tree Protocol (STP) loops, as STP treats the entire bundle as a single logical link.



Condition to become EtherChannel

- Same Ethernet standard
- Same ports speed
- Same Duplex

- Same Native VLAN
- If access port, then all the ports should be in the same VLAN

Types of EtherChannel

Two types

1. Static

2. Dynamic

- PAGP
- LACP

- **LACP (Link Aggregation Control Protocol):** Defined by IEEE 802.3ad, LACP is the vendor-neutral standard for dynamically negotiating these bundles. It exchanges LACP Data Units (LACPDUs) to ensure both sides are capable of forming an aggregation.
- **Channel Group:** The physical ports are assigned to a channel group, which dynamically creates the logical Port-Channel interface in the switch software.
- **PAGP (Port Aggregation Protocol):** A Cisco-proprietary protocol used to dynamically bundle multiple physical ports together.
- *Modes:* **Desirable** (actively initiates PAGP negotiation) and **Auto** (waits to receive PAGP packets).
- **Static (On Mode):** Forces the ports to bundle immediately without sending any negotiation packets (configured via channel-group X mode on). Both sides must be statically configured; if one side misconfigures the speed, duplex, or VLAN settings, it can lead to frame loops or dropped traffic.

1. Maximum Number of Port-Channels per Switch

The total number of logical EtherChannels (e.g., interface Port-channel 1 to X) you can create depends entirely on the Cisco switch model and the IOS version it is running:

- **Access Layer Switches (e.g., Catalyst 2960, 9200):** Typically support up to **48** or **64** EtherChannels.
- **Core/Distribution Switches (e.g., Catalyst 3850, 9300, 9500):** Typically support up to **128** EtherChannels.
- **Nexus Data Center Switches:** Can support up to **256** or more vPCs(Virtual port channel)/EtherChannels depending on the specific line cards and supervisor modules.

2. Maximum Physical Ports per Channel Group

The number of physical ports you can bundle together depends on the negotiation protocol you choose (LACP vs. PAgP):

LACP (IEEE 802.3ad):

You can assign a maximum of **16 physical ports** to a single LACP channel group.

However, only **8 ports can be active** and forwarding traffic at the same time.

The remaining 8 ports are automatically placed into a **hot-standby state**. If one of the active links fails, a standby link will immediately wake up and take its place.

PAgP (Cisco Proprietary):

- You can assign a maximum of **8 physical ports** to a single PAgP channel group.
- All 8 ports are active simultaneously. PAgP does not support a standby pool of backup links like LACP does.

LACP Negotiation Modes

- LACP uses specific modes to determine how a switch negotiates the bundle. A link is only formed if the modes on either side of the connection are compatible.

Local Port Mode	Remote Port Mode	Port-Channel Formed?
Active	Active	Yes
Active	Passive	Yes
Passive	Passive	No (Neither side initiates)

PAgP Mode Compatibility Matrix

For an EtherChannel to form dynamically using PAgP, the modes configured on both sides of the link must be compatible.

Local Port Mode	Remote Port Mode	Port-Channel Formed?
Desirable	Desirable	Yes
Desirable	Auto	Yes
Auto	Desirable	Yes
Auto	Auto	No (<i>Neither side initiates</i>)
On	On	Yes (<i>Static, no negotiation</i>)
On	Desirable / Auto	No (<i>Error state</i>)

Switch# configure terminal

Switch(config)# interface range GigabitEthernet0/1 - 2

Switch(config-if-range)# channel-group 1 mode desirable

Switch(config-if-range)# exit

Switch(config)# interface Port-channel 1

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# end
```

Load Balancing and Configuration Rules

EtherChannel does not strictly balance traffic packet-by-packet (which would cause out-of-order frames). Instead, it uses a hashing algorithm based on packet headers.

- **Hashing Methods:** The switch hashes combinations of Source/Destination MAC addresses, IP addresses, or TCP/UDP port numbers to assign an entire traffic flow to one specific physical link within the bundle.
- **Strict Consistency:** For LACP to bundle links successfully, all physical member ports must be identical. They must have the same speed, duplex settings, Native VLANs, and allowed VLAN ranges.

Layer 2 LACP Port-Channel (Trunk)

A Layer 2 EtherChannel bundles standard switch ports. You configure the physical ports to join the bundle, and then apply your trunking configurations directly to the logical Port-channel interface.

```
Switch(config)# interface range GigabitEthernet0/1 - 2
```

```
Switch(config-if-range)# channel-group 1 mode active
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface Port-channel 1
```

```
Switch(config-if)# switchport trunk encapsulation dot1q
```

```
Switch(config-if)# switchport mode trunk
```

```
Switch(config-if)# exit
```

#show etherchannel summary

Layer 3 LACP Port-Channel (Routed)

A Layer 3 EtherChannel requires you to disable switchport capabilities on the physical interfaces first, bundle them, and then apply an IP address to the logical interface.

```
Switch(config)# interface range GigabitEthernet0/3 - 4
```

```
Switch(config-if-range)# no switchport
```

```
Switch(config-if-range)# channel-group 2 mode active
```

```
Switch(config-if-range)# exit
```

```
Switch(config)# interface Port-channel 2
```

```
Switch(config-if)# no switchport
```

```
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Switch(config-if)# exit
```

Note:

Layer 3 LACP Port-Channels require devices that support both physical link aggregation and network-layer routing. The most common devices used for this configuration include:

Multilayer Switches (Layer 3 Switches)

Cisco Catalyst series (e.g., 3650, 3850, 9300)

Enterprise Routers:

Cisco ISR or ASR series)

Next-Generation Firewalls (NGFW):

Enterprise security appliances (like Fortinet FortiGate, Palo Alto, or Cisco Firepower)

Switch Virtual Interface (SVI)

A Switch Virtual Interface (SVI) is a logical Layer 3 interface configured on a switch. It represents a specific Layer 2 VLAN and allows the switch to route traffic in and out of that VLAN. You configure it in the Cisco IOS by creating an interface for the VLAN (e.g., interface vlan 10).

Primary Functions

- **Default Gateway (Inter-VLAN Routing):** On a Multilayer (Layer 3) switch, SVIs act as the default gateways for the end-hosts (PCs, servers) residing in their respective VLANs. Instead of sending traffic to an external router (like in a "Router-on-a-Stick" topology), the switch can route the traffic between VLANs internally at hardware speeds.
- **Remote Management:** On purely Layer 2 switches (which do not support routing), an SVI is typically configured for the management VLAN (often VLAN 1, or a dedicated management VLAN). Assigning an IP address to this SVI allows administrators to remotely manage the switch using SSH, Telnet, or SNMP.

```
Switch# configure terminal
```

```
ip routing
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# interface vlan 10
```

```
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

SVIs on a Layer 2 Switch

On a strict Layer 2 switch (like an older Cisco Catalyst 2960), the switch only understands MAC addresses for forwarding traffic. It cannot route IP packets between different networks.

- **Primary Purpose: Remote Management.**
- **How it works:** You create an SVI (usually on VLAN 1, or a dedicated Management VLAN) and assign it an IP address. This gives the switch itself a presence on the network, allowing you to ping it or log into it remotely using SSH or Telnet.
- **Limitation:** A pure Layer 2 switch can typically only have **one active SVI** at a time. If you bring up a second SVI, the first one will go down.
- **Routing:** It cannot route traffic. If you want to manage the switch from a completely different network, you must also configure a default gateway on the switch itself using the global command `ip default-gateway <router-ip>`.

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# exit
```

Configure the Management SVI

```
Switch(config)# interface vlan 10
```

```
Switch(config-if)# ip address 192.168.10.10 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

Configure the Default Gateway

```
Switch(config)# ip default-gateway 192.168.10.1
```

```
Switch(config)# end
```

SVIs on a Layer 3 Switch (Multilayer Switch)

A Layer 3 switch (like a Catalyst 3650 or 9300) has specialized hardware (ASICs) that allows it to perform IP routing just like a router, but much faster.

- **Primary Purpose: Inter-VLAN Routing** (and Management).
- **How it works:** You can create **multiple active SVIs**, one for every VLAN on your network (e.g., SVI 10 for VLAN 10, SVI 20 for VLAN 20). You assign an IP address to each SVI, and those IP addresses act as the default gateways for the PCs and servers in those specific VLANs.
- **Routing:** When a PC in VLAN 10 wants to talk to a server in VLAN 20, the traffic hits SVI 10. The Layer 3 switch instantly routes the packet over to SVI 20 and delivers it to the server.
- **Requirement:** To enable this routing capability, you must explicitly turn it on using the ip routing command in global configuration mode.

Switch> enable

Switch# configure terminal

Enable IP Routing Globally

Switch(config)# ip routing

Create VLANs

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# exit
```

Configure SVI for VLAN 10 (Acts as Default Gateway for VLAN 10)

```
Switch(config)# interface vlan 10
```

```
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# exit
```

Configure SVI for VLAN 20 (Acts as Default Gateway for VLAN 20)

```
Switch(config)# interface vlan 20
```

```
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
```

```
Switch(config-if)# no shutdown
```

```
Switch(config-if)# end
```

Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP)

Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP) are Layer 2 neighbor discovery protocols. Because they operate at the data link layer, they function even if Layer 3 routing is broken or IP addresses are misconfigured, making them the most reliable tools for validating physical and logical network documentation.

Essential Data Retrieved

By querying neighbor data, you can verify your diagrams against the actual live environment. Key data points include:

Device ID: The configured hostname of the connected equipment.

Local Interface: The specific port on the device you are currently logged into.

Port ID: The hardware port on the remote neighboring device.

Platform & Capabilities: The hardware model and its role (e.g., router, switch, WLAN controller).

Network Address: The management IP address of the neighbor.

Protocol Comparison

Feature	CDP	LLDP
Standard	Cisco Proprietary	IEEE 802.1AB (Open Standard)
Compatibility	Primarily Cisco devices	Vendor-neutral
Default State	Typically enabled by default on Cisco gear	Often requires manual enablement (lldp run)
Advanced Use	CDPv2 supports rapid PoE negotiation	Extended via LLDP-MED for VoIP endpoints

show cdp neighbors

show lldp neighbors

show cdp neighbors detail

show cdp

show lldp

How to enable cdp?

enable

configure terminal

cdp run

exit

How to enable LLDP?

enable

configure terminal

lldp run

exit

Interface level cdp command

configure terminal

interface GigabitEthernet1/0/1

cdp enable

exit

#show cdp interface GigabitEthernet1/0/1

LLDP allows you to control sending (transmit) and receiving separately on a per-port basis.

configure terminal

interface GigabitEthernet1/0/1

lldp transmit

lldp receive

exit

show lldp interface GigabitEthernet1/0/1

How to disable cdp and lldp

enable

configure terminal

no cdp run

no lldp run

exit

